

4.3.1 Challenges Encountered and Solutions:

Table 1.1: Challenges Encountered and Solutions:

Problem Faced	How I Solved It
Face not detected properly during the exam	Improved camera positioning and lighting conditions
Tab switching not detected properly	Improved JavaScript event handling and tested on different browsers
Automatic warning count increased even after exam submission	Fixed logic to stop monitoring once the exam is submitted
Webcam permission issues	Helped by Chirag Visani sir
Performance issues during real-time monitoring	Optimized code to reduce processing load

Chapter 5
Results and Discussion

5. Results and Discussion

5.1 Overview of Results

After successful implementation, the Smart Online Exam Monitoring System worked as expected. The system was able to conduct online exams securely while monitoring students in real time. Students were able to log in, verify their identity using a webcam, attend exams, and submit answers successfully.

The monitoring module detected activities such as face absence, head movement, and tab switching. Automatic warnings were generated whenever exam rules were violated. When the warning limit was exceeded, the system automatically submitted or terminated the exam. Admin users were able to upload questions, schedule exams, view student activity logs, and check exam results.

5.2 Comparison with Expected Outcomes

The initial objective of the project was to develop a secure and automated online exam system with real-time monitoring. The actual results closely matched the expected outcomes

Table 1.2: Complarision with Expected Outcomes:

Feature	Description	Status
User Registration & Login	Secure login for admin and students	Completed
Admin Exam Creation	Create and schedule exams	Completed
Question Upload	Upload Excel files	Completed
MCQ Extraction	Automatic question extraction	Completed
Student Exam Interface	Exam page with timer	Completed

Face detection	Face check before exam	Completed
Webcam Monitoring	Face presence & head movement detection	Completed
Tab Switch Detection	Detect browser tab switching	Completed
Warning System	Auto warnings for violations	Completed
Auto Exam Submission	Auto-submit after violations	Completed

5.3 Performance Analysis

The performance of the system was analyzed based on speed, accuracy, and stability.

Speed:The system responded quickly during login, exam loading, and submission. Monitoring actions were processed in near real time.

Accuracy:Face detection and tab switching detection worked accurately in most cases. Head movement detection was effective, although minor movements sometimes triggered warnings.

SystemStability:The system remained stable throughout the exam duration without crashes. Data was stored correctly in the database.

5.4 Limitations of the Project

- The system depends heavily on webcam quality and lighting conditions
- Minor head movements may cause false warnings
- Internet connectivity issues can affect real-time monitoring
- The system is designed mainly for desktop or laptop browsers
- Advanced AI-based behavior analysis is not included
- These limitations can be improved in future versions of the system.

5.5 Discussion on Findings

The results of the project show that automated online exam monitoring is possible using simple and lightweight technologies. Webcam-based monitoring and browser activity tracking proved effective in reducing cheating.

Face detection and tab switching detection worked well and helped maintain exam discipline. Automatic warnings and exam termination reduced the need for human supervision. However, sensitivity tuning is required to avoid false warnings due to small movements.

The project successfully demonstrated that a cost-effective and automated system can improve the reliability of online examinations. With further improvements, the system can be made more accurate and scalable for real-world use.

Outputs:

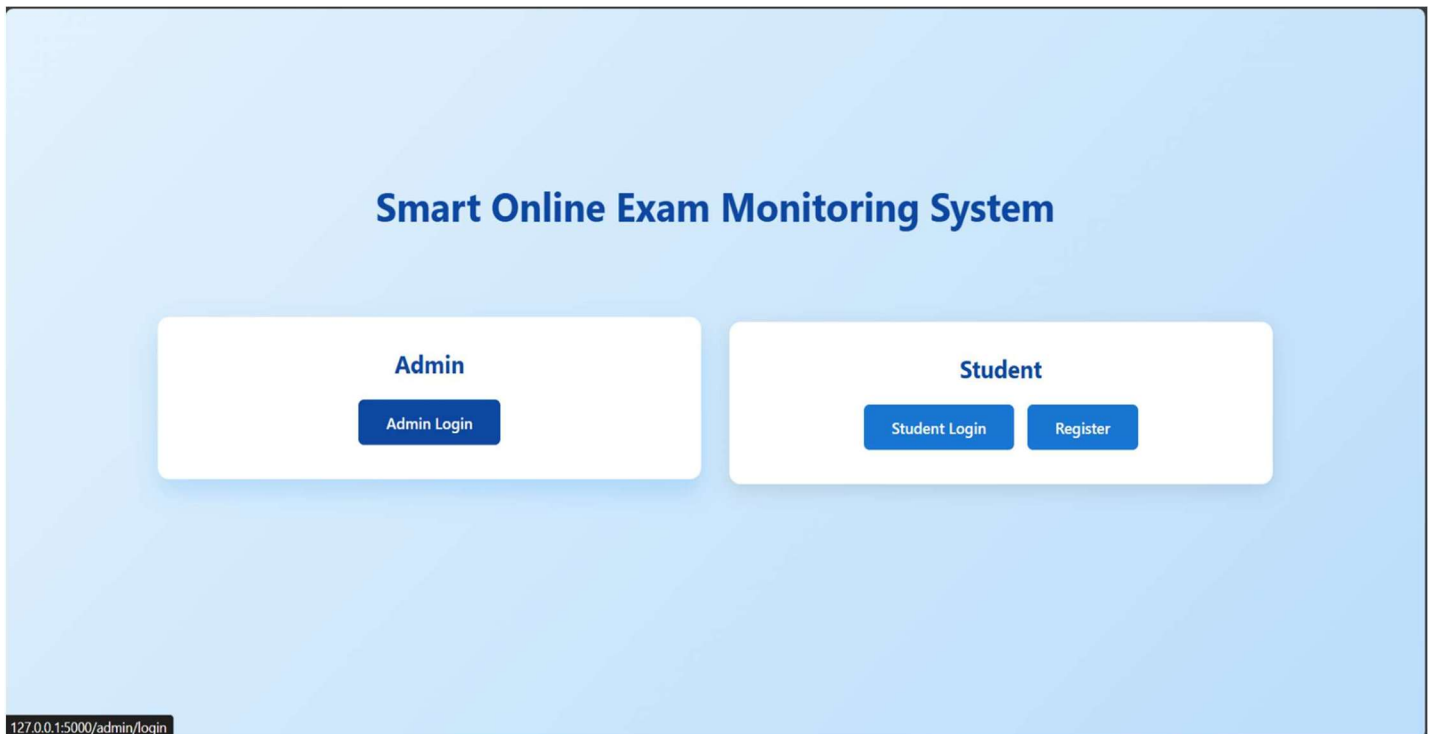
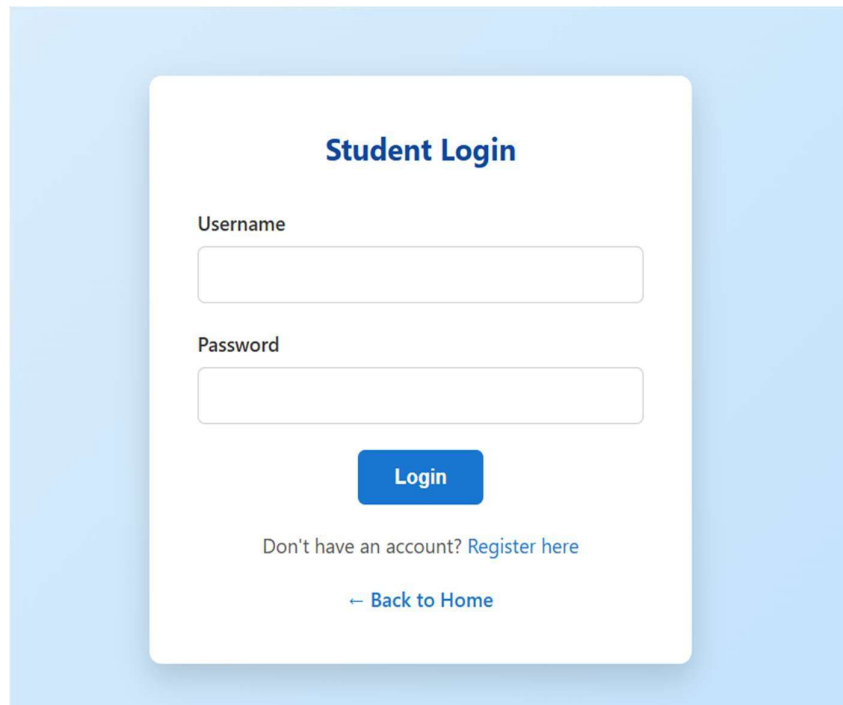


Figure 3.1.8 Main Screen

The image shows a 'Student Login' form centered on a light blue background. The form is a white rounded rectangle. At the top, the title 'Student Login' is in bold blue text. Below it are two input fields: 'Username' and 'Password', each with a light gray border. A blue 'Login' button is positioned below the password field. At the bottom of the form, there is a link 'Don't have an account? Register here' and a link '← Back to Home' in blue text.

Student Login

Username

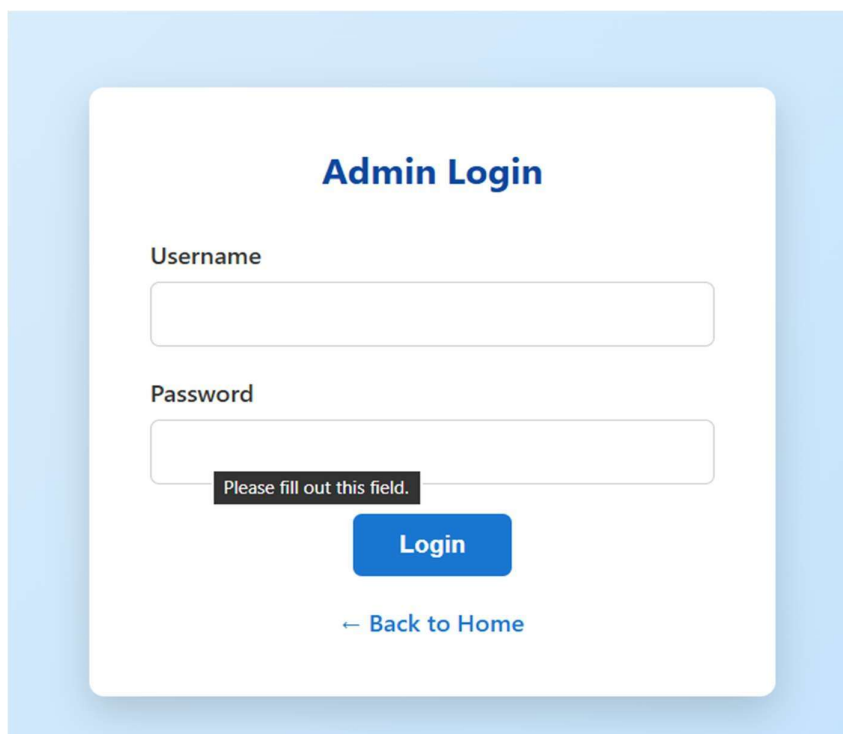
Password

Login

Don't have an account? [Register here](#)

[← Back to Home](#)

Figure 3.1.9 Student Login

The image shows an 'Admin Login' form centered on a light blue background. The form is a white rounded rectangle. At the top, the title 'Admin Login' is in bold blue text. Below it are two input fields: 'Username' and 'Password', each with a light gray border. A blue 'Login' button is positioned below the password field. A black error message box with the text 'Please fill out this field.' is overlaid on the bottom of the password input field. At the bottom of the form, there is a link '← Back to Home' in blue text.

Admin Login

Username

Password

Please fill out this field.

Login

[← Back to Home](#)

Figure 3.2 Admin Login

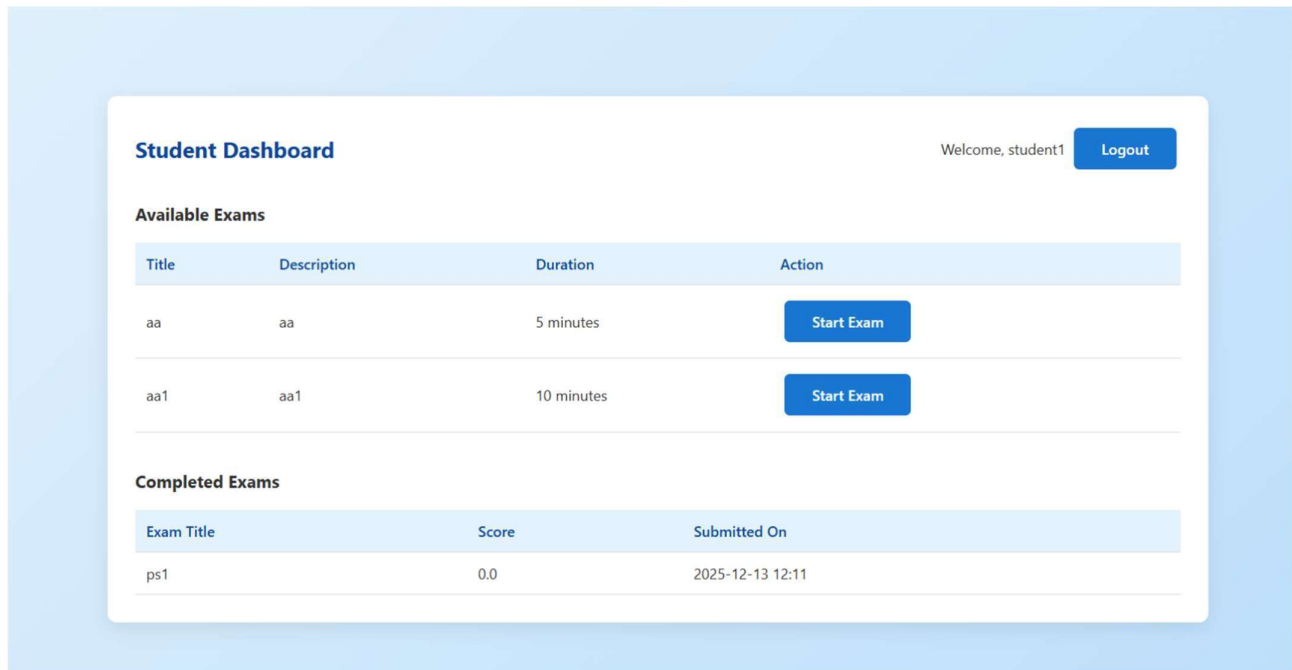


Figure 3.2.1 Student Dashboard

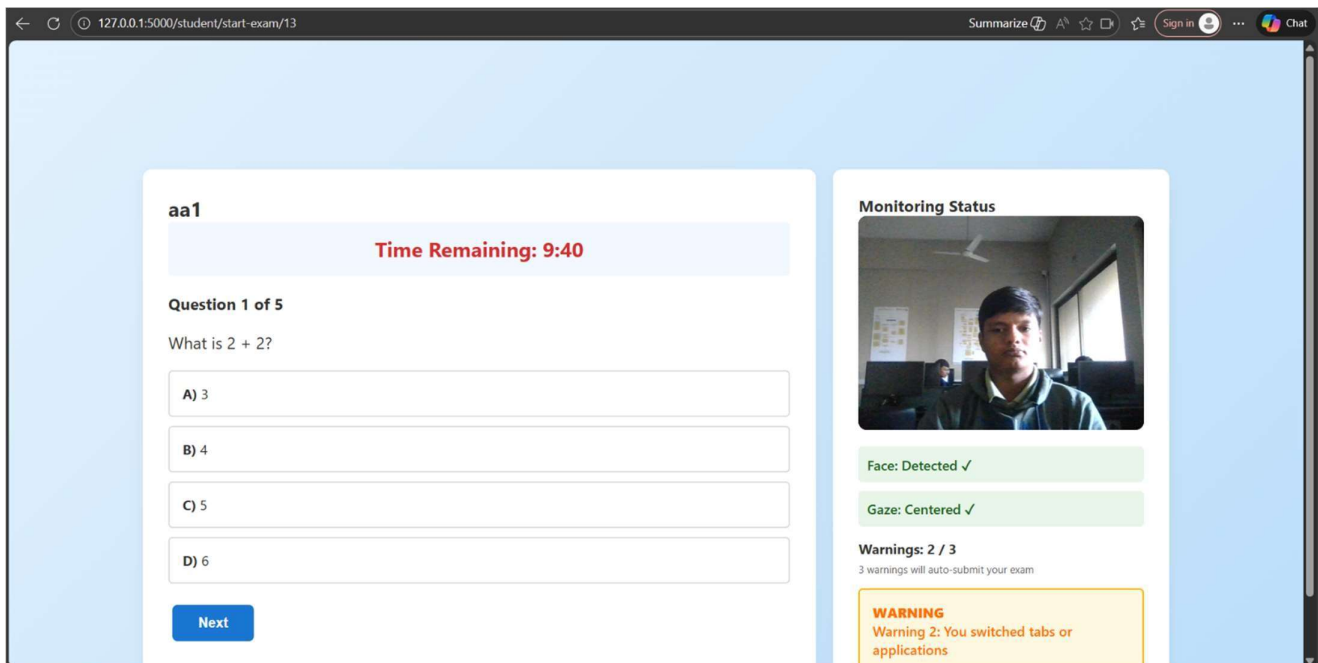


Figure 3.2.2 Live exam view

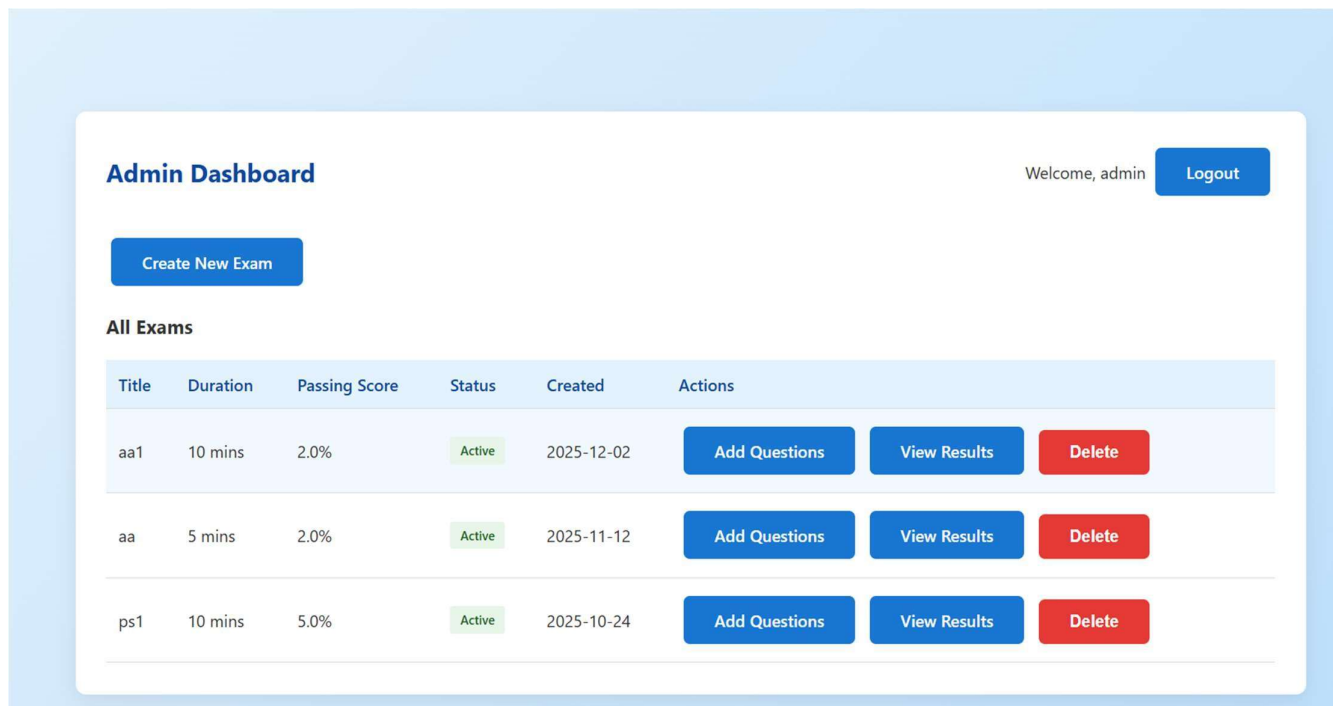


Figure 3.2.3 Admin Dashboard

The 'Create New Exam' form includes the following fields and controls:

- Exam Title ***: A text input field.
- Description**: A text area with a tooltip that says 'Please fill out this field.'
- Duration (minutes) ***: A text input field.
- Passing Score (%) ***: A text input field.
- Create Exam**: A blue button to submit the form.
- Cancel**: A blue button to cancel the form.

Figure 3.2.4 Create new exam page

Upload Questions for: aa1

Excel File Format Instructions

Your Excel file must have the following columns (starting from row 2):

Question	Option A	Option B	Option C	Option D	Answer
Capital of France?	London	Paris	Rome	Berlin	B

Note: The "Answer" column must contain A, B, C, or D.

Upload Excel File (.xlsx or .xls only):

No file chosen

Figure 3.2.5 Question upload page

Monitoring Logs

Exam Details

Exam: ps1

Student: Test Student (student1)

Score: 0.0 / 10

Warnings: 3

Status: terminated

Activity Timeline

Total time taken: 34 seconds (0.6 min)

Time	Face Detected	Gaze Direction	Head Pose	Warning	Details
06:41:02	-	-	-	-	Exam Started
06:41:02	-	-	-	Warning	Warning 1 issued
06:41:16	-	-	-	Warning	Warning 2 issued
06:41:29	-	-	-	Warning	Warning 3 issued
06:41:36	-	-	-	-	Exam Ended

Figure 3.2.6 Student logs

Results for: ps1

Back to Dashboard

Student	Score	Total Marks	Percentage	Warnings	Status	Submitted At	Actions
Test Student (student1)	0.0	10	0.0%	3	Terminated	2025-12-13 12:11	View Logs

Figure 3.2.7 Student marks view

sqliteviewer.app/#/exam_system.db/table/users/

exam_system.db > users

Search tables...

Reset Filters

Records: 2

Search 2 records...

id	username	password	email	full_name	role	created_at
1	admin	script:32768:8:1\$18...	...	System Administrator	admin	2025-10-19 13:55:26
2	student1	script:32768:8:1\$1b...	student1@test.com	Test Student	student	2025-12-13 06:38:45

sqliteviewer.app/#/exam_system.db/table/exams/

exam_system.db > exams

Search tables...

Reset Filters

Records: 3

Search 3 records...

id	title	description	duration_min	passing_score	is_active	created_at
1	ps1	ps1	10	5	1	2025-10-24 10:52:04
2	aa	aa	5	2	1	2025-11-12 08:08:00
3	aa1	aa1	10	2	1	2025-12-02 06:15:38

sqliteviewer.app/#/exam_system.db/table/questions/

exam_system.db > questions

Search tables...

Reset Filters

Records: 25

Search 25 records...

id	exam_id	question_text	option_a	option_b	option_c	option_d	correct_answer
1	11	2 What is the capital ...	Berlin	Madrid	Paris	Rome	C
2	12	2 Which planet is kno...	Venus	Mars	Jupiter	Saturn	B
3	13	2 What does HTTP st...	Hyper Text Transfer ...	High Tech Transfer ...	Home Tool Transfer...	Hyperlink Text Prog...	A
4	14	2 Which programmin...	C++	JavaScript	Swift	Kotlin	B
5	15	2 What is 2 + 2?	3	4	5	6	B
6	16	2 What does CSS sta...	Cascading Style She...	Computer Style Sys...	Creative Style Softw...	Colorful Site Sheets	A
7	17	2 Which company cre...	Google	Microsoft	Facebook	Amazon	C
8	18	2 What is the largest ...	Atlantic Ocean	Indian Ocean	Arctic Ocean	Pacific Ocean	D
9	19	2 What year did Worl...	1943	1944	1945	1946	C
10	20	2 Which of these is N...	List	Dictionary	Array	Tuple	C
11	26	4 What is 2 + 2?	3	4	5	6	B
12	27	4 Capital of France?	London	Paris	Berlin	Madrid	B
13	28	4 Which is a program...	Python	Snake	Apple	Banana	A
14	29	4 What color is the sk...	Blue	Green	Yellow	Red	A
15	30	4 HTML stands for?	Hyper Trainer MarkL...	Hyper Text Markup ...	Hyper Text Marketi...	High Text Machine ...	B

Figure 3.2.8 Databases view

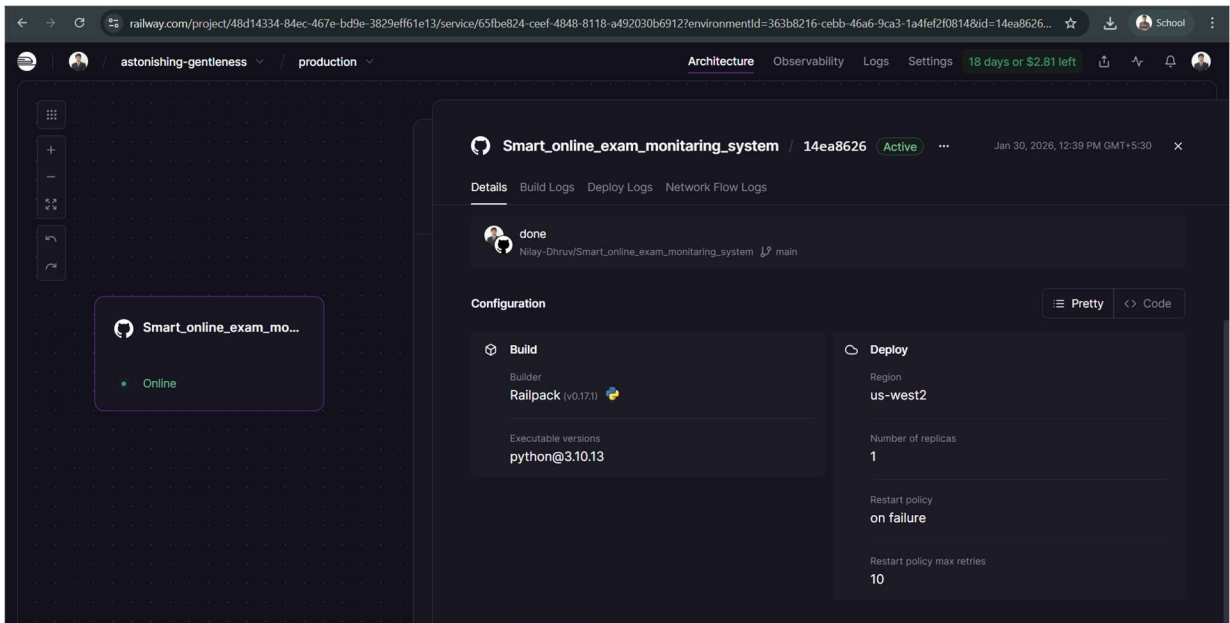


Figure 3.2.9 Deployment view

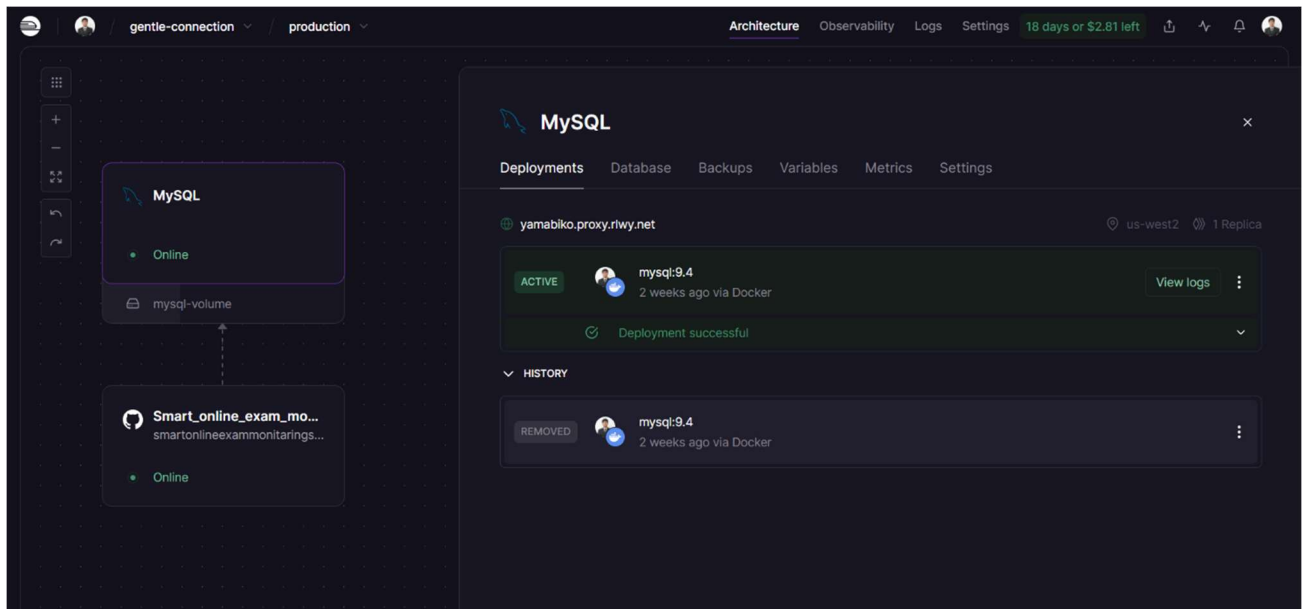


Figure 3.3.0 Deployment view

Chapter 6
Conclusion and Future Work

6. Conclusion and Future Work

6.1 Conclusion

The Smart Online Exam Monitoring System was successfully designed and implemented to conduct online examinations in a secure and controlled manner. The system addressed major problems found in traditional online exams, such as lack of supervision and increased chances of cheating. By using webcam monitoring and browser activity tracking, the system ensured better discipline and fairness during examinations.

The project achieved its main objectives, including face verification before the exam, real-time monitoring during the exam, detection of suspicious activities like looking away and tab switching, and automatic warning generation. The admin module allowed easy exam management, including question upload, exam scheduling, and result analysis.

Overall, the system proved to be reliable, cost-effective, and easy to use. It reduced the need for human invigilators and improved the credibility of online exam results. The project successfully demonstrated that automated monitoring can enhance the integrity of online examinations.

6.2 Future Work

Although the system works effectively, there is scope for further improvement. In future, the following enhancements can be added:

- Improve face and head movement detection accuracy using advanced AI models
- Add voice detection to identify external help during exams
- Support mobile devices and tablets
- Include live admin monitoring during exams
- Improve handling of poor internet connectivity
- Add cloud database support for large-scale deployment
- Integrate identity verification using ID cards or biometric methods

Chapter 7
References

7. References

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